



10.5.1 Additional homework ¶

Homework 10.5.1.1. You may want to do a new "git pull" to update directory `Assignments`.

In `Assignments/Week10/matlab` you will find the files

- `Givens_rotation.m`: A function that computes a Givens' rotation from a 2×1 vector `x`.
- `Francis_Step.m`: A function that performs a Francis Implicit QR Step with a tridiagonal matrix T (stored as the diagonal and subdiagonal of `T`).
- `Test_Francis_Step.m`: A very rudimentary script that performs a few calls to the function `Francis_Step`. Notice that our criteria for the routine being correct is that the matrix retains the correct eigenvalues.

With this,

1. Investigate the convergence of the $(m, m - 1)$ element of matrix `T1`.
2. Write a function

```
function T = Spectral_Decomposition_Lambda( T )
```

That returns Λ such that $T = Q\Lambda Q^T$ is the Spectral Decomposition of T . The input matrix T is a tridiagonal matrix where only the lower triangular part of the matrix is stored in the diagonal and first subdiagonal of array `T`. The diagonal matrix Λ is returned in `T`. The upper triangular part of the array should not change values. You are encouraged to call the function `Francis_Step` from the function `Spectral_Decomposition_Lambda`. Obviously, you need to incorporate deflation in your implementation. How to handle the final 2×2 matrix is an interesting question... (You may use the matlab function `eig` for this.)